

Towards Decentralized Experimental Implementation of Distance-based Formation Controllers

Tonmoy Sarker¹, Marcio de Queiroz²



Dept. Mechanical and Industrial Eng., Louisiana State University, Baton Rouge, LA, USA

¹Graduate Student, ²Faculty Member

INTRODUCTION

- A well-known approach for the multi-agent formation problem is **distance-based control**, which involves regulating a set of inter-agent distances to values prescribed by a desired shape.
- In this approach, sensing, algorithm processing, and communication can be done from hardware onboard the agents in a fully **decentralized** manner.
- An indoor robotic experimental testbed called **TIGER Square** has been developed at the LSU iCORE Lab to enable multi-agent system behavior analysis in centralized and decentralized modes.
- This work seeks to demonstrate no deterioration occurs in the formation performance when the system operates with decentralized processing.

TIGER SQUARE TESTBED

Arena

An enclosed 1.5 x 1.5 m platform with two overhead cameras, one for centralized robot localization, another for recording experiments.

Control Station

Computer with MATLAB software package. WiFi router for communication with the robots.

TIGERBots

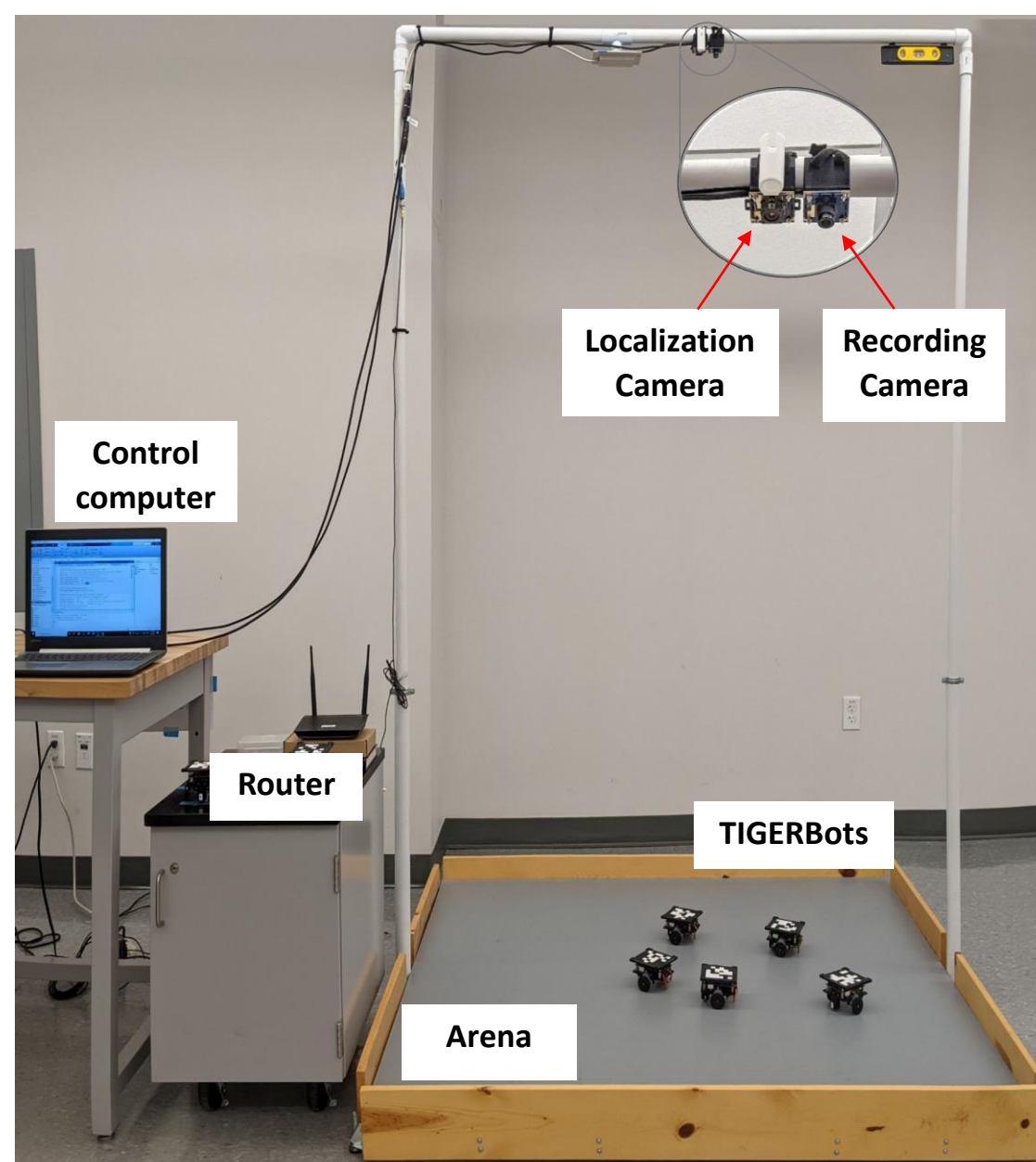
A fleet of ten robots.

- Each robot is 9x9x9.5cm in size with a maximum speed of 3.5 cm/s.
- Robot's main board is controlled by ESP8266 NodeMCU Wi-Fi chip. Each robot is powered by a 4.2 V Li-Po battery.
- For decentralized sensing, each robot is equipped with a Teensy 3.2 controlled infrared (IR) sensor board with 8 photodiodes and 16 LEDs (360 degree sensing) to measure relative range and bearing from neighboring robots.



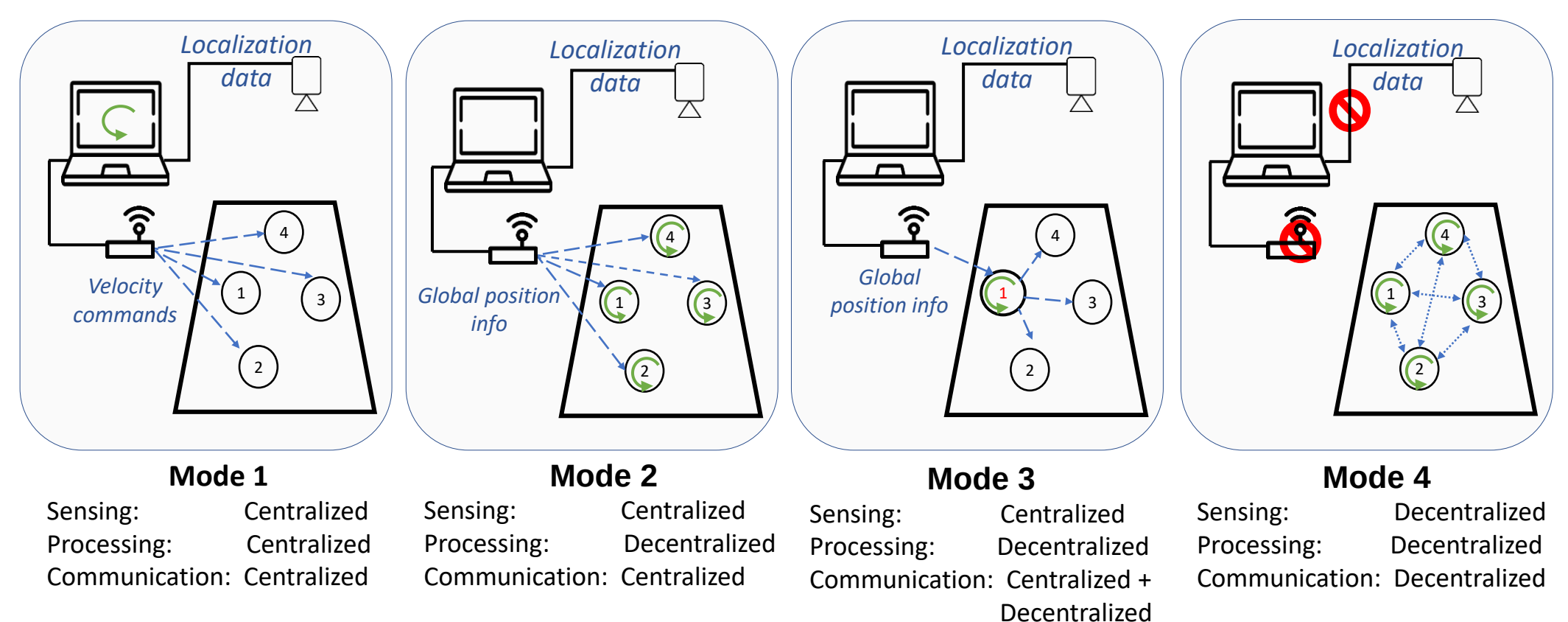
Main board

IR Sensor board



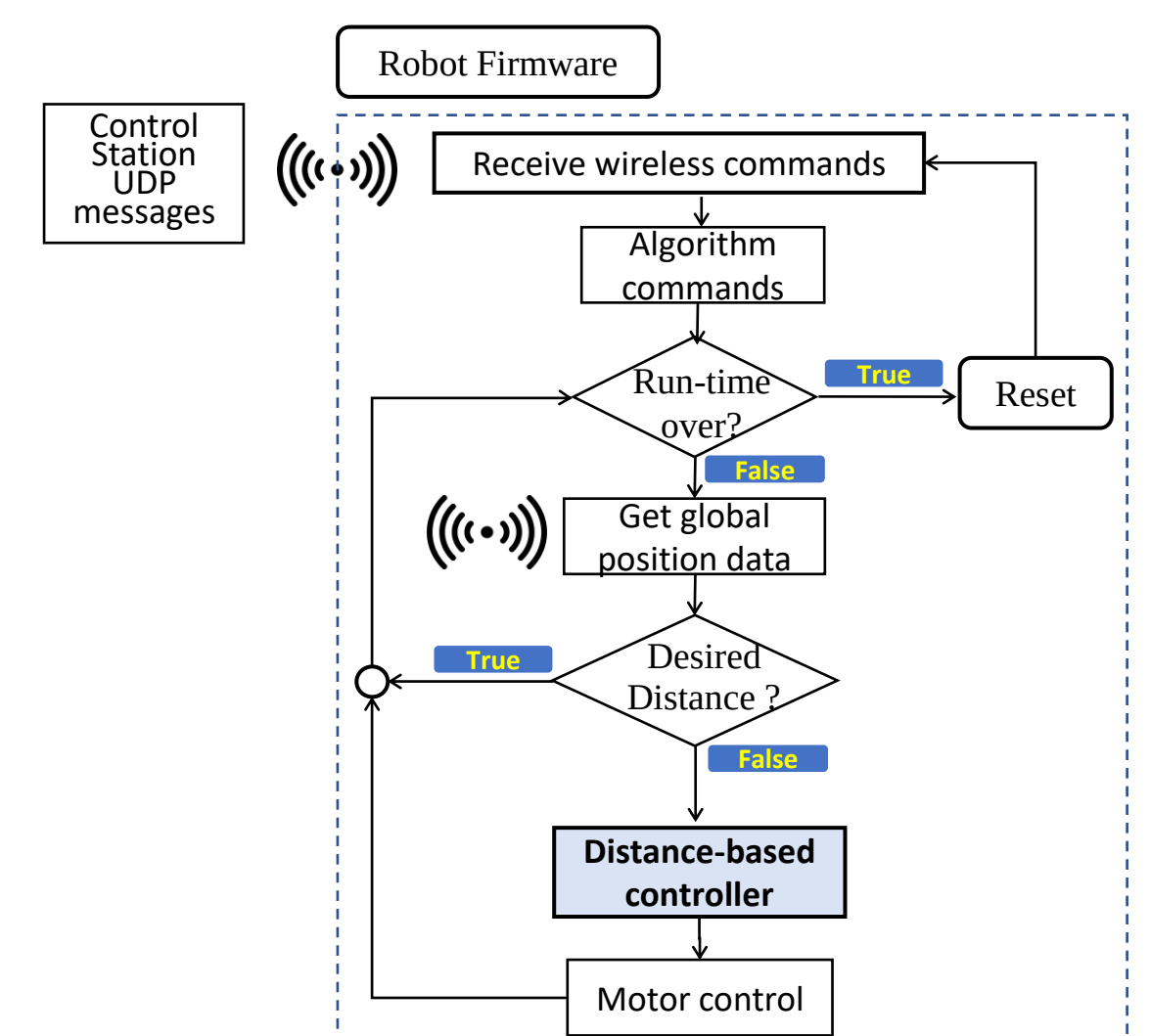
OPERATING MODES

- The testbed is currently being refined to operate in 4 operating modes based on combination of centralized or decentralized sensing, processing, and communication.



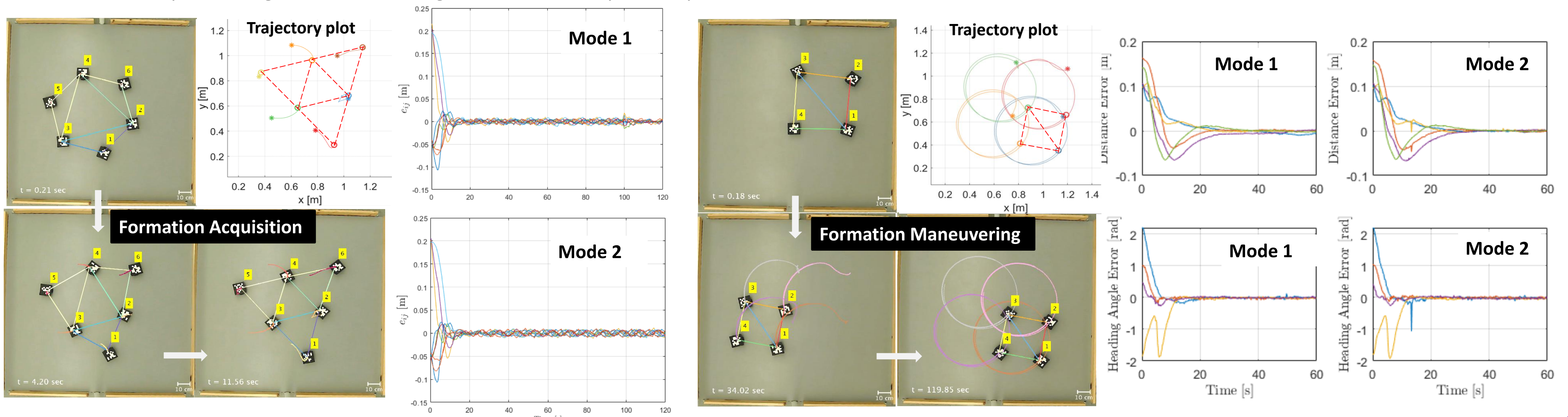
DECENTRALIZED PROCESSING

- Decentralized processing is done on robot's onboard NodeMCU microcontroller.
- Each robot computes its own control algorithm using C++ with Arduino and NodeMCU libraries upon receiving camera pose information from the control station.



EXPERIMENTAL RESULTS

- Two formation experiments were conducted: **Formation acquisition**, which seeks to control the agents to acquire and maintain a predefined geometric shape in space; and **Formation maneuvering**, which requires the agents to simultaneously acquire a formation and maneuver cohesively along a predefined trajectory.
- Agents' convergence times in Modes 1 and 2 in both experiments do not show any significant differences.
- Decentralized processing of the control algorithms has comparable performance to centralized controllers.



FUTURE WORK

- Refine the testbed subsystems to switch seamlessly between centralized and decentralized coordination schemes while online.
- In order to improve the current decentralized sensing method, new techniques including onboard image processing and machine learning could be implemented alongside currently used IR sensor boards.